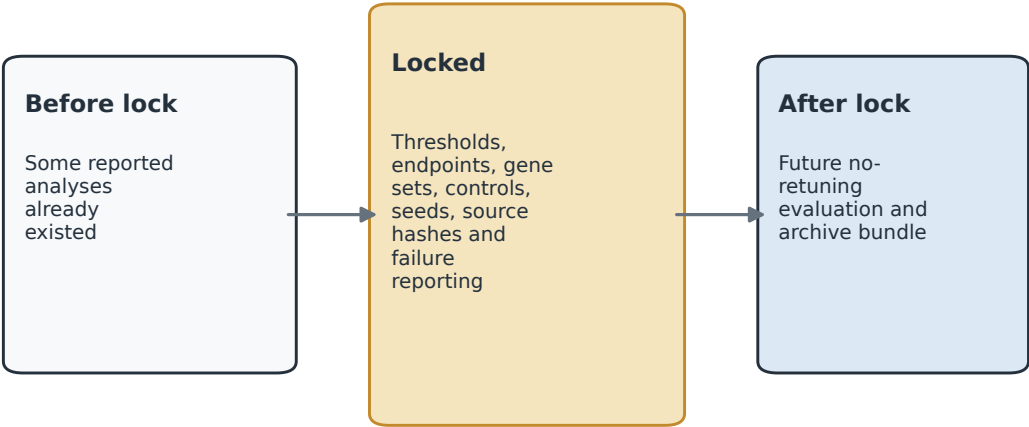


Frozen benchmark design

A. Retrospective no-retuning lock

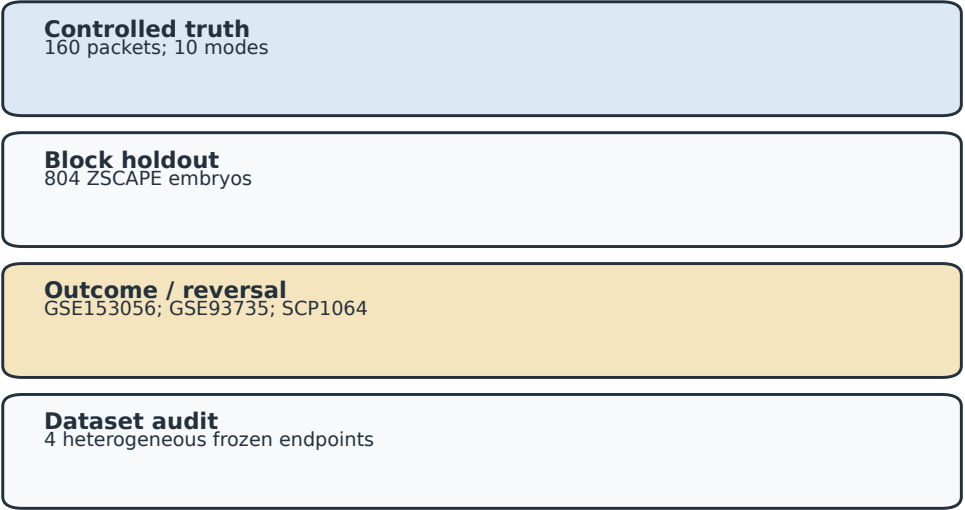


Not prospective preregistration; no new post-freeze dataset yet

C. Truth source and inferential unit

Stratum	Truth / readout	Unit
Packets	controlled event + artifact labels	packet
ZSCAPE	full-data event object	embryo
GSE153056	PD-L1 protein	perturbation
GSE93735	intervention reversal	sample contrast
SCP1064	RNA-protein + shuffles	block / guide / target

B. Evaluation strata



D. Fair comparator hierarchy

Missing TED fields are reported as coverage, not native-task errors

